

Name: _____ Class: 16S _____ Reg Number: _____



MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination
Higher 2

Chemistry

9729/01

Paper 1 Multiple-Choice Questions

22 September 2017

1 hour

Additional Materials: *Data Booklet*
 OMR Answer Sheet

INSTRUCTIONS TO CANDIDATES

Write your name, class and register number in the spaces provided at the top of this page.

There are **thirty** questions in this section. Answer **all** questions. For each question, there are four possible answers labelled **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the OMR answer sheet.

Read very carefully the instructions on the OMR answer sheet.

You are advised to fill in the OMR Answer Sheet as you go along; no additional time will be given for the transfer of answers once the examination has ended.

Use of OMR Answer Sheet

Ensure you have written your name, class register number and class on the OMR Answer Sheet. Use a **2B** pencil to shade your answers on the OMR sheet; erase any mistakes cleanly. Multiple shaded answers to a question will not be accepted.

For shading of class register number on the **OMR sheet**, please follow the given examples:

If your register number is **1**, then shade **01** in the index number column.

If your register number is **21**, then shade **21** in the index number column.

This document consists of **14** printed pages

- 1 The use of Data Booklet is relevant to this question.

Scientists have been comparing the stability of isotopes using the neutron:proton ratio. Nuclei are considered to be stable if the neutron:proton ratio is between 1:1 to 1.5:1. Stable nuclei within these ratios are known to be in the '*band of stability*'.

Which one of these elements do not lie within the '*band of stability*' ?

- A ^{91}Zr B ^{133}Cs C ^{165}Ho D ^{209}Bi

- 2 The information on two particles, Q^+ and R^+ , are given in the table below.

particles	number of electrons	number of neutrons	angle of deflection in an electric field
Q^+	20	25	2.0°
R^+		20	2.3°

Which of the following pairs of particles are isoelectronic?

- A Q and R
 B Q and R^+
 C Q^+ and R
 D Q^+ and R^+

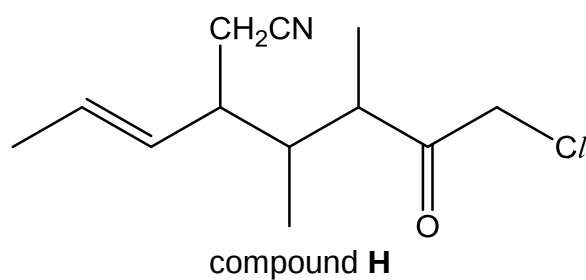
- 3 Three substances **D**, **E** and **F** have the following physical properties.

substance	melting point / $^\circ\text{C}$	boiling point / $^\circ\text{C}$	electrical conductivity in	
			solid state	liquid state
D	1538	2862	good	good
E	2852	3600	poor	good
F	3550	3825	good	not known

What could be the identities of **D**, **E** and **F**?

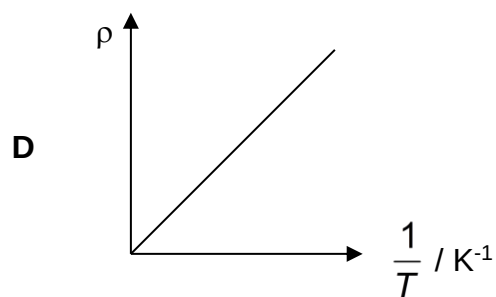
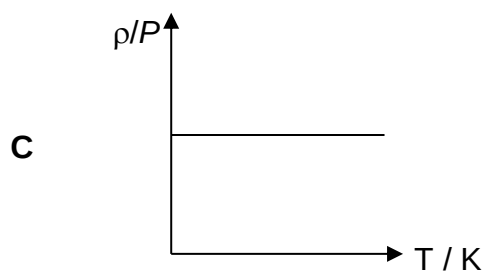
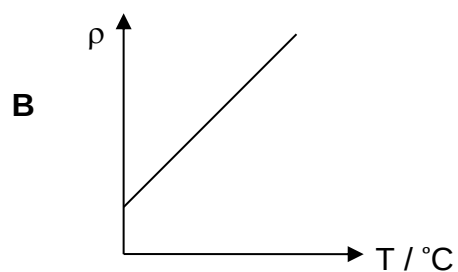
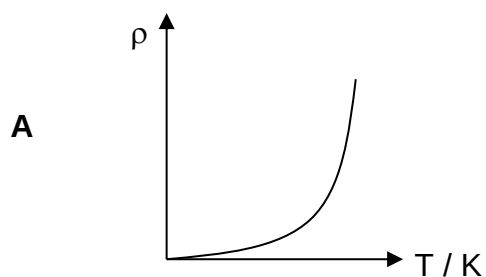
	D	E	F
A	Fe	NaCl	SiO_2
B	Fe	MgO	C (graphite)
C	NaCl	MgO	C (diamond)
D	Fe	SiO_2	C (graphite)

- 4 What is the number of sp , sp^2 and sp^3 hybridised carbons present in compound **H**?



	sp hybridised C	sp^2 hybridised C	sp^3 hybridised C
A	0	3	7
B	1	3	7
C	1	3	8
D	1	4	8

- 5 Which of the following graphs show the correct relationship for an ideal gas under constant pressure, P ?



6 In which reaction does ammonia behave as a Lewis base?

- 1 $\text{HCl} + \text{NH}_3 \longrightarrow \text{NH}_4\text{Cl}$
- 2 $\text{NaH} + \text{NH}_3 \longrightarrow \text{NaNH}_2 + \text{H}_2$
- 3 $\text{Zn}^{2+} + 4\text{NH}_3 \longrightarrow [\text{Zn}(\text{NH}_3)_4]^{2+}$

- A 1 only
 B 3 only
 C 1 and 3 only
 D 1, 2 and 3

7 A chloride and an oxide of the elements in the third period of the Periodic Table are dissolved in two separate portions of water to form aqueous solutions.

Both of the resulting solutions can be used to dissolve Al_2O_3 but only one of the two can be used to dissolve SiO_2 .

Which of the following could be the chloride and the oxide used?

- A NaCl MgO
 B MgCl_2 SO_3
 C SiCl_4 Na_2O
 D PCl_5 P_4O_{10}

8 Three solutions containing the halogens X_2 , Y_2 and Z_2 were added to three aqueous solutions of X^- , Y^- and Z^- ions separately. The experimental results are shown in the following table.

	X^-	Y^-	Z^-
X_2	No reaction	No reaction	X^- formed
Y_2	Y^- formed	No reaction	Y^- formed
Z_2	No reaction	No reaction	No reaction

Which of the following statements is correct?

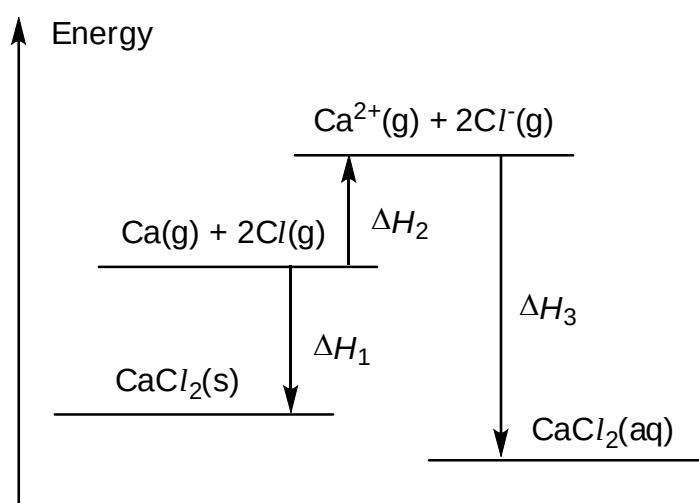
- A Y_2 is a stronger oxidising agent than X_2 .
 B X_2 is a weaker oxidising agent than Z_2 .
 C Y^- is a stronger reducing agent than Z^- .
 D Z^- is a weaker reducing agent than X^- .

- 9 **W** is a compound containing only Fe, C and H. When 1.888 g of **W** was subjected to complete combustion, 4.47 g of CO_2 and 0.914 g of H_2O were formed.

What is the empirical formula of **W**?

- A FeCH
- B FeC_2H
- C $\text{FeC}_{10}\text{H}_{10}$
- D $\text{Fe}_2\text{C}_5\text{H}_2$

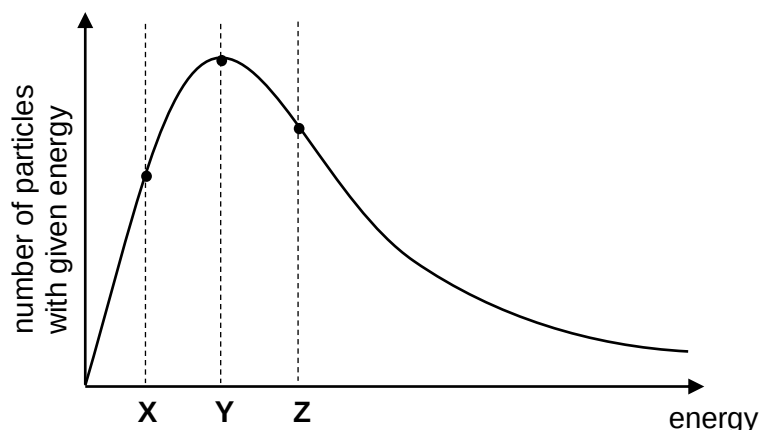
- 10 The energy level diagram shown represents the dissolution of CaCl_2 in water.



Which of the following statements about the diagram is correct?

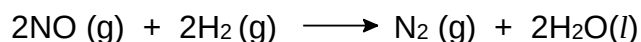
- A The enthalpy change of formation of CaCl_2 is given by ΔH_1 .
- B The enthalpy change of dissolving CaCl_2 is given by $\Delta H_1 + \Delta H_2 + \Delta H_3$.
- C The sum of the first and second ionisation energies of Ca is given by ΔH_2 .
- D The sum of the enthalpy change of hydration of Ca^{2+} and Cl^- is given by ΔH_3 .

- 11** The Maxwell–Boltzmann distribution for a gas at constant temperature is shown below.

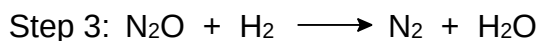
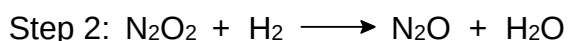


How would the number of particles change at the kinetic energies X, Y and Z when temperature is lowered by 5 °C.

- | | X | Y | Z |
|----------|--------|--------|--------|
| A | lower | lower | higher |
| B | lower | higher | higher |
| C | higher | lower | lower |
| D | higher | higher | lower |
- 12** The reaction of nitric oxide with hydrogen can be monitored by the initial rate method and the overall kinetics is found to be third order.



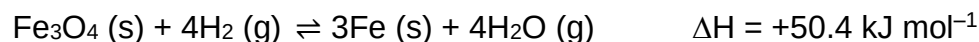
A proposed mechanism of the reaction is given below.



What is the slow step in the mechanism?

- A** Step 1
B Step 2
C Step 3
D No conclusion can be drawn from the mechanism

- 13** Fe_3O_4 is found naturally in the mineral magnetite. It can be reduced to iron metal using hydrogen according to the following equilibrium.



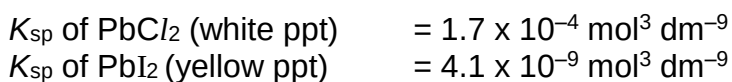
Which of the following statements are correct about the reaction?

- 1 A reduction in the total pressure will increase the production of Fe.
- 2 The numerical values of K_c and K_p are equal.
- 3 An increase in temperature will increase the value of K_c .

- A** 1, 2 and 3
B 1 and 2 only
C 2 and 3 only
D 3 only

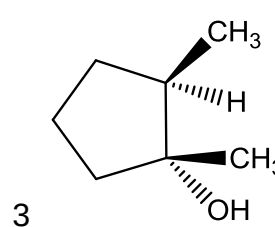
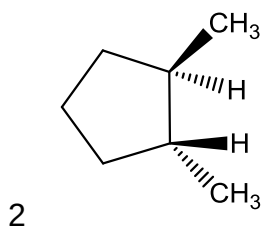
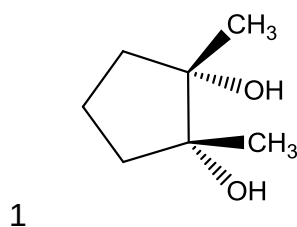
- 14** An equal volume of 0.1 mol dm^{-3} $\text{Pb}(\text{NO}_3)_2$ is added to a mixture containing 0.1 mol dm^{-3} of NaCl and 0.1 mol dm^{-3} NaI.

What is the most probable observation?



- A** A white and yellow ppt is formed.
B A yellow ppt is formed.
C A white ppt is formed.
D No ppt is formed.

- 15** Which molecules are optically active?



- A** 3 only
B 1 and 2 only
C 2 and 3 only
D 1, 2 and 3 only

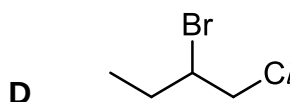
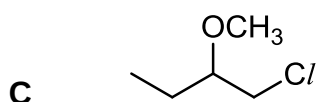
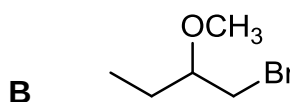
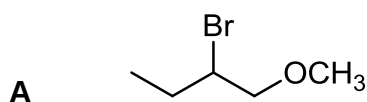
- 16** One mole of methylcyclobutane reacts with one mole of halogen in the presence of ultra-violet light.

Which statements are true about this reaction?

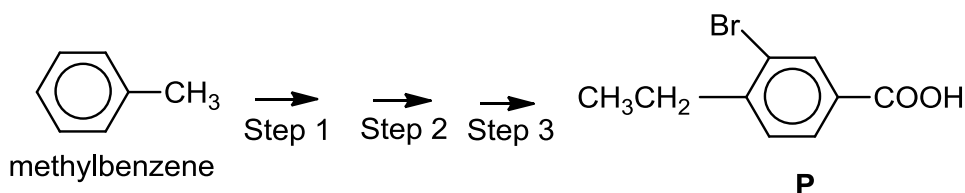
- 1 There are only four possible mono-substituted products.
- 2 Hydrogen is a side product formed in the reaction.
- 3 Homolytic fission occurs only in the initiation step.

- A** 1 only
B 1 and 2 only
C 1 and 3 only
D 2 and 3 only

- 17** Which of the following is the major product when but-1-ene reacts with BrCl in pure methanol?



- 18** Compound **P** can be synthesised from methylbenzene as shown below.

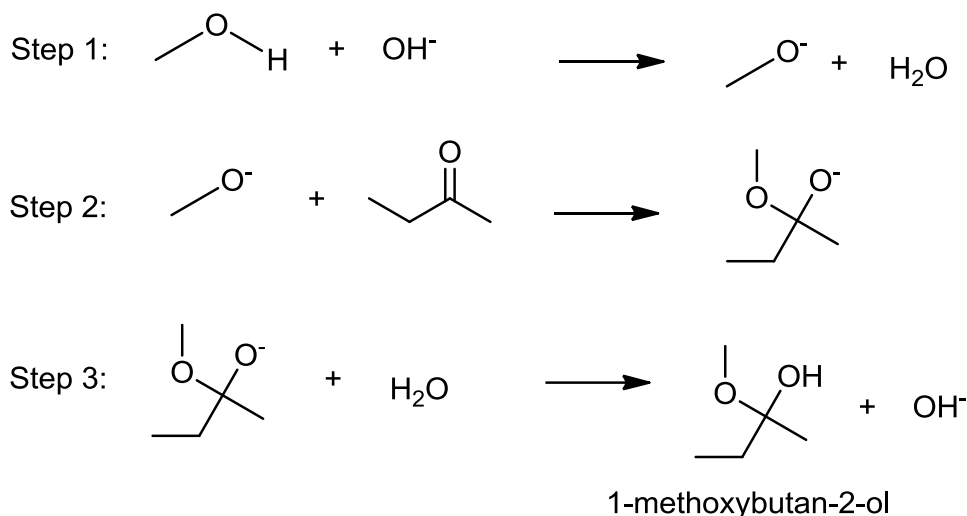


Which of the following could be a sequence for converting methylbenzene to compound **P**?

	step 1	step 2	step 3
A	$\text{CH}_3\text{CH}_2\text{Cl}$, AlCl_3 , r.t.p.	Br_2 in AlBr_3 , in the dark	KMnO_4 , dilute H_2SO_4 , heat
B	KMnO_4 , dilute H_2SO_4 , heat	Br_2 in AlBr_3 , heat	$\text{CH}_3\text{CH}_2\text{Cl}$, AlCl_3 , r.t.p
C	Br_2 in AlBr_3 , heat	$\text{CH}_3\text{CH}_2\text{Cl}$, AlCl_3 , r.t.p	$\text{K}_2\text{Cr}_2\text{O}_7$, dilute H_2SO_4 , heat
D	$\text{K}_2\text{Cr}_2\text{O}_7$, dilute H_2SO_4 , heat	$\text{CH}_3\text{CH}_2\text{Cl}$, AlCl_3 , r.t.p.	Br_2 in AlBr_3 , in the dark

- 19** 1-methoxybutan-2-ol is an organic compound which has an alcohol and an ether (R–O–R) attached to the same carbon atom. It is formed when butan-2-one reacts with methanol in the presence of a catalyst.

The reaction follows the mechanism below.



Which of the following statements about the reaction is **not** correct?

- A** Methanol acts as a Lewis acid in step 1.
 - B** Step 2 is a nucleophilic addition reaction across the C=O bond.
 - C** Methanol can be used in step 3 instead of water.
 - D** KOH can be used as a catalyst for this reaction.
- 20** 0.5 g of each of the following compounds was heated with NaOH (aq).



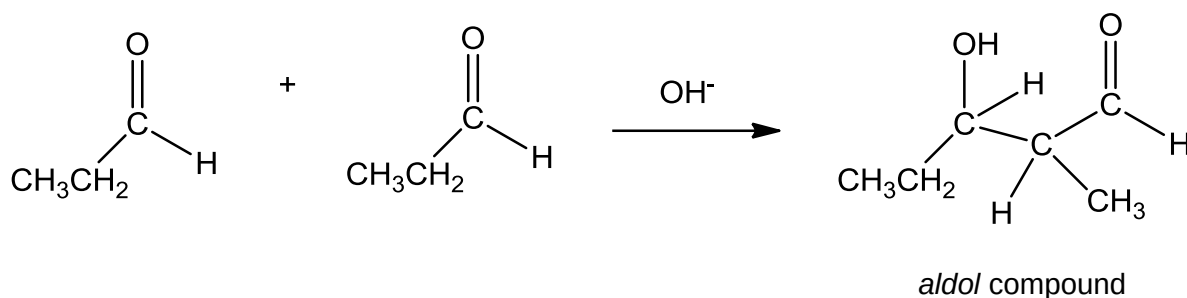
Each of the mixtures was then acidified with dilute nitric acid, followed by the subsequent addition of AgNO_3 (aq).

Which statement is correct?

- A** The reaction with $\text{C}_6\text{H}_5\text{Br}$ gave a cream precipitate.
- B** The reaction with $\text{CH}_3\text{CH}_2\text{I}$ gave a precipitate that dissolved completely in dilute aqueous ammonia.
- C** The rate of reaction with $\text{CH}_2=\text{CHCl}$ would be the slowest.
- D** The reaction with CH_3COBr gave the largest mass of precipitate.

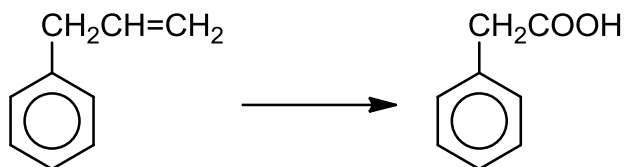
- 21** In the presence of a strong base, aldehydes and ketones may undergo addition reaction to form an *aldo* compound (hydroxycarbonyl).

A reaction of propanal with sodium hydroxide to form an *aldo* compound is as shown below.



Which product can be formed when sodium hydroxide is added to a mixture of butanal, $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$, and propanone, CH_3COCH_3 ?

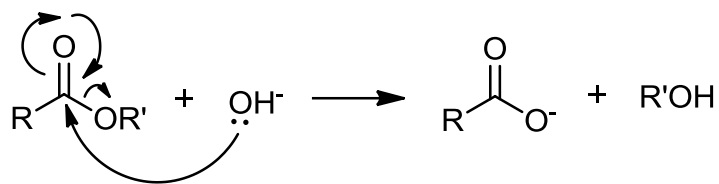
- A** $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_2\text{CHO}$
B $(\text{CH}_3)_2\text{C}(\text{OH})\text{CH}_2\text{CH}_2\text{CH}_2\text{CHO}$
C $(\text{CH}_3\text{CH}_2)_2\text{C}(\text{OH})\text{CH}_2\text{COCH}_3$
D $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}(\text{OH})\text{CH}_2\text{COCH}_3$
- 22** Alkenes can undergo oxidation to form carboxylic acids. The following conversion occurs in a multiple-step reaction.



Which of the following reagents is involved in the above conversion?

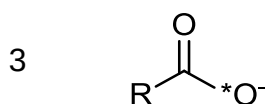
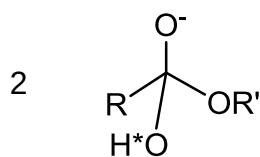
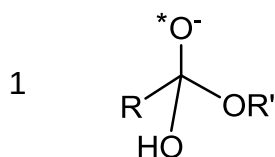
- A** concentrated H_2SO_4
B acidified $\text{K}_2\text{Cr}_2\text{O}_7$
C acidified KMnO_4
D alkaline I_2

23 Esters can be hydrolysed under basic conditions according to the following mechanism.



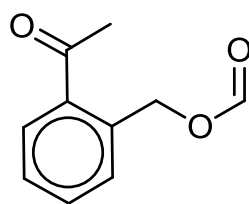
In an isotope labelling experiment, oxygen-18-labeled hydroxide, $^*\text{OH}^-$, is used.

Which of the following are produced as an intermediate or product?

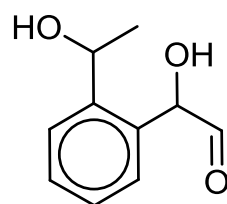


- A** 1 and 2 only
B 1 and 4 only
C 2 and 3 only
D 3 and 4 only

24 Compounds **V** and **W** have the following structures.



compound **V**

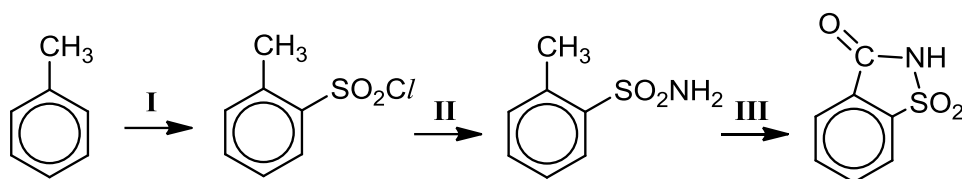


compound **W**

Which of the following reagents can be used to distinguish between compounds **V** and **W**?

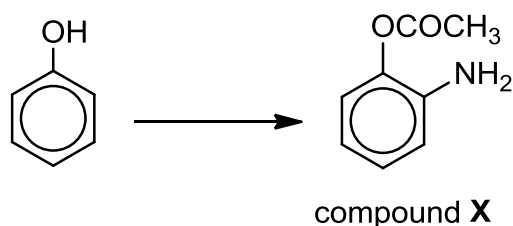
- A** iodine and aqueous sodium hydroxide
B 2,4-dinitrophenylhydrazine
C acidified aqueous potassium dichromate(VI)
D aqueous silver nitrate and aqueous ammonia

- 25 Saccharin is an artificial sweetener which can be synthesised from methylbenzene through a series of reactions below.



Which of the following statements is correct about the above synthesis?

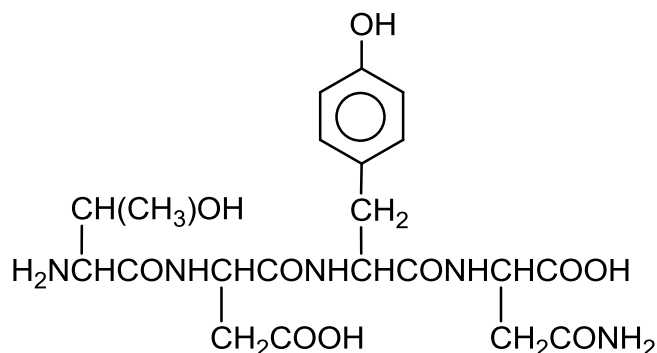
- A Reaction I involves methylbenzene as an electrophile.
 B Reaction II is a nucleophilic substitution reaction.
 C Reaction III does not involve oxidation.
 D Reaction III does not involve the elimination of water.
- 26 Phenols are widely used in household products and as intermediates for industrial synthesis. Compound X can be produced from phenol in three steps.



What are the reagents required in this three-step synthesis?

	stage 1	stage 2	stage 3
A	concentrated HNO_3 , concentrated H_2SO_4	NaBH_4	CH_3COCl
B	aqueous HNO_3	Sn , concentrated HCl followed by NaOH (aq)	CH_3COCl
C	aqueous HNO_3	CH_3COCl	Sn , concentrated HCl , followed by NaOH (aq)
D	aqueous HNO_3	NaBH_4	CH_3COCl

- 27 Compound Y is formed from the partial hydrolysis of a hormone molecule.



Compound Y

Which of the following statement is **incorrect**?

- A Prolonged heating of compound Y with dilute NaOH produces **three** carbon-containing products.
- B Prolonged heating of compound Y with dilute NaOH liberates an alkaline gas.
- C One mole of Na_2CO_3 is needed for complete reaction with one mole of compound Y.
- D Two moles of Na is needed for complete reaction with one mole of compound Y.

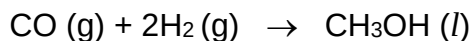
- 28 Use of the Data Booklet is relevant to this question.

A galvanic cell consists of a Mg^{2+}/Mg half-cell and a Cu^{2+}/Cu half-cell.

Which of the following statements is correct?

- A Reducing the size of the magnesium electrode decreases the e.m.f of the cell.
- B Increasing the temperature has no effect on the e.m.f. of the cell.
- C Adding aqueous sodium hydroxide to the Cu^{2+}/Cu half-cell increases the e.m.f. of the cell.
- D Adding water to the the Mg^{2+}/Mg half-cell increases the e.m.f. of the cell.

- 29** Methanol can be synthesised according to the reaction below. Cr_2O_3 is used as a catalyst to increase the rate of this reaction.



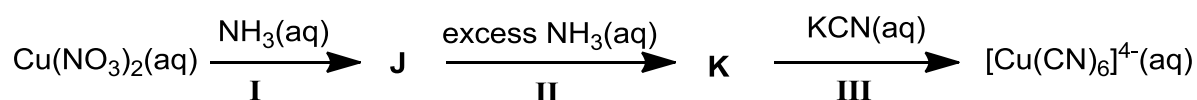
Which of the following statements are correct about the synthesis of methanol?

- 1 Cr_2O_3 is used as a catalyst because Cr can exhibit variable oxidation states in its compounds.
- 2 Cr_2O_3 lowers the activation energy by using its partially filled 3d orbitals for the adsorption of reactant molecules.
- 3 There is a decrease in the oxidation number of carbon.

- A** 1, 2 and 3
B 1 and 2 only
C 2 and 3 only
D 1 only

- 30** A reaction scheme starting from aqueous copper(II) nitrate solution is shown below.

Both **J** and **K** are copper-containing species.



Which of the following statements are correct about the above reaction scheme?

- 1 One of the reactions involves a redox reaction.
- 2 CN^- is a stronger ligand than NH_3 .
- 3 Precipitation occurs in step **I**.

- A** 1, 2 and 3
B 2 and 3 only
C 1 and 2 only
D 2 only

End of Paper 1